🔐 Why Information Security?

Information Security (InfoSec) is the practice of protecting information by mitigating risks to its **confidentiality, integrity, and availability** (the **CIA triad**). In the digital era, where data is a crucial asset, safeguarding it from unauthorized access, modification, or destruction is vital for business continuity, privacy, and trust.

Key Drivers for Information Security:

1. **Digital Transformation** – More organizations are moving online, increasing their exposure to threats.

2. **Data Sensitivity** – Financial, personal, and health-related data need protection from misuse.

3. **Compliance Requirements** – Regulations like GDPR, HIPAA, PCI-DSS require stringent InfoSec measures.

4. **Business Reputation** – Breaches result in loss of customer trust and market value.

5. **Cybercrime Growth** – With increasing sophistication in attack methods, robust security is essential.

🔐 Security: The Money Factor Involved

Implementing security isn't just about technology—it also affects an organization's **finances** in major ways.

1. Cost of a Breach:

- **IBM 2023 Cost of a Data Breach Report**: Average global cost = **$4.45 million**.
- Includes legal fines, downtime, forensic investigation, and customer churn.

2. Investment in Security:

- Organizations now budget **5-15% of IT spend** on cybersecurity.
- Cost includes: firewalls, IDS/IPS, antivirus, SIEM tools, audits, awareness training, and expert staff.

3. ROI of Security:

- Though expensive upfront, good security saves **millions** in avoided breach costs.
- **Risk reduction** = better business resilience and investor confidence.

🌐 Internet Statistics – Security Perspective

Understanding global internet usage helps assess **threat exposure**:

2024 Internet Stats (Security Relevance):

- **Over 5.4 billion** internet users.
- **90%+ businesses** use cloud services.
-

400 million+ new malware samples annually (AV-TEST).

- **Phishing attacks** increased by **30% year-over-year**.

- **IoT devices** surpassing 20 billion – increasing attack surface.

Implication:

- More users and devices = more **attack vectors**.

- Global connectivity requires **global security posture**.

- Massive data flow needs **monitoring, encryption, and control**.

Vulnerability, Threat, and Risk

Understanding these three core terms is essential in InfoSec:

Term	Definition	Example
Vulnerability	A weakness in a system that can be exploited	Unpatched software
Threat	A potential cause of an unwanted incident	Hacker attempting to access data
Risk	The probability of a threat exploiting a vulnerability and the resulting impact	The risk of losing customer data due to SQL injection

Relationship:

Risk = Threat × Vulnerability × Impact

Reducing risk involves:

- Reducing vulnerabilities (patching)

- Reducing threats (firewalls, IDS)
- Reducing impact (backups, DR plans)

🔷 QoS (Quality of Service)

Though not traditionally an InfoSec concept, **QoS** intersects with security in ensuring **availability**, a key part of the CIA triad.

What is QoS?

- Set of technologies to manage **network traffic** and ensure **reliable data transmission**.
- Prioritizes critical services (e.g., VoIP, real-time apps).

Security Link:

- Helps mitigate **DoS/DDoS attacks** by prioritizing legitimate traffic.
- QoS tools can detect **abnormal traffic behavior**, aiding in security detection.

📌 Summary of Core Concepts

Concept	Summary
Why InfoSec?	To protect data and ensure privacy, trust, and compliance.
Money Factor	High cost of breaches → investment in security saves money long-term.
Internet Stats	Increased connectivity = higher exposure to attacks.
V/T/R	Core triad of risk assessment in cybersecurity.
QoS	Ensures network reliability; supports availability and mitigates disruptions.